

Cybersecurity Internship – Week 1 Documentation

1. Application Setup

A mock web application was downloaded from GitHub and configured using:

- Node.js
- Express.js
- MySQL

Steps Performed

- Installed dependencies using npm install
- Configured MySQL database
- Imported SQL file
- Started application using npm start
- Tested Register and Login pages

2. Vulnerability Assessment

OWASP ZAP Scan

OWASP ZAP was used to scan: <http://localhost:3000>

Vulnerabilities Detected

- SQL Injection
- Path Traversal
- Missing CSRF Protection
- Missing Security Headers
- Cookie Security Issues

XSS Testing

Test Payload: <script>alert('XSS')</script>

Result

Application accepted script input but popup did not execute.
Possible weak input sanitization detected.

SQL Injection Testing

Test Payload: ' OR '1'='1

Result

Browser validation partially blocked the payload, but OWASP ZAP detected possible SQL Injection vulnerabilities.

Weak Password Storage

Passwords stored in database were hashed using MD5.

3. Areas of Improvement

- Replace MD5 with bcrypt
- Add input validation
- Use Helmet.js
- Implement JWT authentication
- Add CSRF protection
- Improve password policy